Reflect with us - Week 8

This assignment will not be graded and is only for practice.
1 point
Choose the set of correct options.

☐ Suppose $\beta = \{v_1, v_2, \ldots, v_n\}$β = {v1, v2, … , vn} is an orthogonal basis of an inner product space V V . If there exists some $v \in V$v ∈ V , such that $\langle v, v_i \rangle = 0$⟨v, vi⟩ = 0 for all $i = 1, 2, \ldots, n$ i = 1, 2, … , n, then $v = 0$v = 0.

☐ There exists an orthonormal basis for R^n Rn with the standard inner product.

☐ If P_W PW denotes the linear transformation which projects the vectors of an inner product space VV to a subspace WW of VV , then $range(P_W) \cap null\ space(P_W) = \{0\}$range(PW) ∩ null space(PW) = {0}, where 00 denotes the zero vector of VV .

☐ $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$[10−11] cannot represent a matrix corresponding to some projection.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Suppose $\beta = \{v_1, v_2, \ldots, v_n\}$β = {v1, v2, … , vn} is an orthogonal basis of an inner product space V V . If there exists some $v \in V$v ∈ V , such that $\langle v, v_i \rangle = 0$⟨v, vi⟩ = 0 for all $i = 1, 2, \ldots, n$ i = 1, 2, … , n, then $v = 0$v = 0.
There exists an orthonormal basis for R^n Rn with the standard inner product.
If P_W PW denotes the linear transformation which projects the vectors of an inner product space VV to a subspace WW of VV , then $range(P_W) \cap null\ space(P_W) = \{0\}$range(PW) ∩ null space(PW) = {0}, where 00 denotes the zero vector of VV .
$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$[10−11] cannot represent a matrix corresponding to some projection.

Solution:

Option 1:Option 1:
$\beta = \{v_1, v_2, \ldots, v_n\}$β = {v1, v2, … , vn} is an orthonormal basis of the inner product space VV .
Step 1:Step 1: As $v \in V$v ∈ V , we can express vv as a linear combination of the vectors $v_1, v_2, \ldots, v_n$ v1, v2, … , vn as follows:
$v = \alpha_1 v_1 + \ldots + \alpha_n v_n$ v = α1v1 + … + αnvn, where $\alpha_1, \ldots, \alpha_n \in R$α1, … , αn ∈ R
Taking inner product of vv with v_1 v1, we have $\langle v, v_1 \rangle = \langle \alpha_1 v_1 + \ldots + \alpha_n v_n, v_1 \rangle$⟨v, v1⟩ = ⟨α1v1 + … + αnvn, v1⟩
Recall two properties of inner products on VV :
i) $\langle u_1 + u_2, u_3 \rangle = \langle u_1, u_3 \rangle + \langle u_2, u_3 \rangle$⟨u1 + u2, u3⟩ = ⟨u1, u3⟩ + ⟨u2, u3⟩ for any vectors $u_1, u_2, u_3 \in V$u1, u2, u3 ∈ V .
ii) $\langle cu_1, u_2 \rangle = c\langle u_1, u_2 \rangle$⟨cu1, u2⟩ = c⟨u1, u2⟩ for any vectors $u_1, u_2 \in V$u1, u2 ∈ V and c ∈ R.
Using these properties we have,

$\langle v, v_1 \rangle = \alpha_1 \langle v_1, v_1 \rangle + \alpha_2 \langle v_2, v_1 \rangle + \ \ldots \ + \alpha_n \langle v_n, v_1 \rangle$⟨v, v1⟩ = α1⟨v1, v1⟩ + α2⟨v2, v1⟩ + … + αn ⟨vn, v1⟩.

What is the value of $\langle v_1, v_1 \rangle$⟨v1, v1⟩?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

What is the value of $\langle v_2, v_1 \rangle$⟨v2, v1⟩?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

What is the value of $\langle v_i, v_1 \rangle$⟨vi, v1⟩, for any $i \in \{2, 3 \ldots, n\}$i ∈ {2, 3 … , n}?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Feedback:Feedback: Recall the definition of an orthonormal basis and also observe that ββ is an orthonormal basis.

If $\langle v, v_1 \rangle = 0$⟨v, v1⟩ = 0, then what is the value of α_1 α1?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Feedback:Feedback: Observe that $\langle v, v_1 \rangle = \alpha_1 \langle v, v1 \rangle$⟨v, v1⟩ = α1
Similarly we can conclude that, $\alpha_i = 0,$ for each $i = 1, 2, \ \ldots, n$αi = 0, for each i = 1, 2, … , n
Hence, $v = 0$v = 0.
Option 2:Option 2: Let $\gamma = \{v_1, v_2, \ \ldots, v_n\}$γ = {v1, v2, … , vn} be a given ordered basis of a vector space. The steps in the process are described below to yield an orthonormal basis $\beta = \{u_1, u_2, \ \ldots, u_n\}$ β = {u1, u2, … , un}.
Step-1 $w_1 = v_1$ w1 = v1 , and $u_1 = \frac{w_1}{\|w_1\|}$u1 = ‖w1‖w1.
Step-2 $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$ w2 = v2 − ⟨v2, u1⟩u1 and $u_2 = \frac{w_2}{\|w_2\|}$u2 = ‖w2‖w2.
Step-3 $w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$ w3 = v3 − ⟨v3, u1⟩u1 − ⟨v3, u2⟩u2 and $u_3 = \frac{w_3}{\|w_3\|}$u3 = ‖w3‖w3.

⋮ ⋮
Step-i $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$ wi = vi − ∑j=1i−1⟨vi, uj⟩uj and $u_i = \frac{w_i}{\|w_i\|}$ui = ‖wi‖wi.
⋮ ⋮
Step-n $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$ wn = vn − ∑j=1n−1⟨vn, uj⟩uj and $u_n = \frac{w_n}{\|w_n\|}$un = ‖wn‖wn.

Hence, Option 2 is correct.

Option 3:Option 3: The projection of a vector to a subspace:The projection of a vector to a subspace: Let VV be an inner product space, $v \in V$v ∈ V and $W \subseteq V$W ⊆ V be a subspace. Then the projection of vv onto WW is the vector in WW, denoted by $P_W(v)$PW (v), computed as follows:

Find an orthonormal basis $\{v_1, v_2, \ldots, v_n\}$ for W. Find an orthonormal basis {v1, v2 , … , vn} for W.

Then $P_W(v) = \sum_{i=1}^{n} \langle v, v_i \rangle v_i$ Then PW (v) = ∑i=1n⟨v, vi⟩vi

Step 1:Step 1:

$$\begin{aligned} P_W^2(v) &= P_W(P_W(v)) \\ &= P_W\left(\sum_{i=1}^{n} \langle v, v_i \rangle v_i\right) \\ &= \sum_{i=1}^{n} P_W(\langle v, v_i \rangle v_i) \\ &= \sum_{i=1}^{n} \langle v, v_i \rangle P_W(v_i) \end{aligned}$$ PW 2(v) = PW (PW (v)) = PW (i=1∑n⟨v, vi⟩vi) = i=1∑n PW

(⟨v, vi⟩vi) = i=1∑n⟨v, vi⟩PW (vi)
1 point
At this point can you identify $P_W(v_i)$PW (vi)?
◯
v_i vi
◯ 0
No, the answer is incorrect.
Score: 0
Accepted Answers:
v_i vi

Feedback:Feedback: As v_i vi is itself in WW, the projection of it onto WW is nothing but the vector v_i vi itself. On applying the definition, we get, $P_W(v_i) = \sum_{j=1}^{n} \langle v_i, v_j \rangle v_j = v_i$ PW (vi) = ∑j=1n⟨vi , vj⟩vj = vi as $\langle v_i, v_j \rangle = 0$⟨vi, vj⟩ = 0 when $i \neq j$i = j and $\langle v_i, v_i \rangle = 1$⟨vi, vi⟩ = 1.
Hence we have

$$P_W^2(v) = \sum_{i=1}^{n} \langle v, v_i \rangle P_W(v_i) = \sum_{i=1}^{n} \langle v, v_i \rangle v_i = P_W(v)$$PW 2(v) = ∑i=1n⟨v, vi⟩ PW (vi) = ∑i=1n⟨v, vi⟩vi = PW (v)

So we can conclude $P_W^2 = P_W$ PW 2 = PW .
Step 2:Step 2: Let $v \in range(P_W) \cap null\ space(P_W)$v ∈ range(PW) ∩ null space(PW). So $v \in range(P_W)$v ∈ range(PW) and $v \in null\ space(P_W)$v ∈ null space(PW).
As $v \in range(P_W)$v ∈ range(PW) there exists $v' \in V$v′ ∈ V such that $P_W(v') = v$PW (v′) = v, and as $v \in null\ space(P_W)$v ∈ null space(PW), we have $P_W(v) = 0$PW (v) = 0.

$$
\begin{aligned}
P_W(v') &= v \\
P_W^2(v') &= P_W(P_W(v')) \\
P_W(v') &= P_W(v) [\text{as, } P_W^2 = P_W] \\
P_W(v') &= 0 \\
v &= 0
\end{aligned}
$$

(v′)) = PW (v)[as, PW 2 = PW] = 0 = 0
Hence $range(P_W) \cap null\ space(P_W) = \{0\}$range(PW) ∩ null space(PW) = {0}.
Option 4:Option 4: If a matrix AA represents a projection, then A^2 A2 must be AA.
Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$A = [10−11].

1 point
Is $A^2 = A$A2 = A?
- ○ Yes
- ○ No

No, the answer is incorrect.
Score: 0
Accepted Answers:
No

Hence, $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$[10−11] cannot represent a matrix corresponding to some projection.

Check Answers
Your score is: 0/7